Computational Maths Sheet 8

1) $O(n^2)$

2) Let $f \in C^{\infty}([a,b], \mathbb{R})$. Assume that Newton's method from a fixed initial guess $x_0 \in [a,b]$ gives a sequence of iterates which remain in $[a,b]$ and converges to $\alpha \in [a,b]$

a) Show that Newton's method is at least quadratically convergent to α if $f'(\alpha) \neq 0$.

error $= e_n = x_n - \alpha$, $x_n = e_n + \alpha$, e_n is small

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}$$

$$f(x_n) = f(\alpha) + f'(\alpha)e_n + \frac{1}{2}f''(\alpha)e_n^2 + O(e_n^3)$$

$$= f'(\alpha)e_n + \frac{1}{2}f''(\alpha)e_n^2 + O(e_n^3)$$

$$f'(x_n) = f'(\alpha) + f''(\alpha)e_n + O(e_n^2)$$

$$e_{n+1} + \alpha = e_n + \alpha - \frac{f'(\alpha)e_n + \frac{1}{2}f''(\alpha)e_n^2 + O(e_n^3)}{f'(\alpha) + f''(\alpha)e_n + O(e_n^2)}$$

$$e_{n+1} = \frac{\frac{1}{2}f''(\alpha)e_n^2 + e_n(O(e_n^2)) - (O(e_n^3))}{f'(\alpha) + f''(\alpha)e_n + O(e_n^2)}$$

$$e_{n+1} \approx \frac{1}{2}\frac{f''(\alpha)}{f'(\alpha)}e_n^2 + O(e_n^3)$$

$|e_{n+1}| \leq C|e_n|^2$, so quadratically convergent

b) Show that Newton's method is linearly convergent if $f'(\alpha)=0$, $f''(\alpha)\neq 0$

$$e_n = x_n - \alpha, \quad x_n = e_n + \alpha$$

$$e_{n+1} = e_n - \frac{f(\alpha+e_n)}{f'(\alpha+e_n)}$$

$$e_{n+1} = e_n - \frac{\frac{1}{2}f''(\alpha)e_n^2 + O(e_n^3)}{f''(\alpha)e_n + O(e_n)^2}$$

$$\simeq \frac{1}{2}e_n + O(e_n^2)$$

So linear converge with rate $1/2$

c) How does the rate of convergence vary if $f'(\alpha)=0$, $f''(\alpha)=0$, $f'''(\alpha)\neq 0$

$$f(x_n) = \frac{1}{6}f'''(\alpha)e_n^3 + O(e_n^4)$$
$$f'(x_n) = \frac{1}{2}f'''(\alpha)e_n^2 + O(e_n^3)$$

$$e_{n+1} = e_n - \frac{\frac{1}{6}f'''(\alpha)e_n^3 + O(e_n^4)}{\frac{1}{2}f'''(\alpha)e_n^2 + O(e_n^3)}$$

$$e_{n+1} = \frac{2}{3}e_n + O(e_n^2)$$

linear convergence with rate $2/3$

d) $f'(\alpha)=0$, $f''(\alpha)=0$

$$e_{n+1} = e_n - \frac{f'(\alpha)e_n + \frac{1}{6}f'''(\alpha)e_n^3 + O(e_n^4)}{f'(\alpha) + \frac{1}{2}f''(\alpha)e_n^2 + O(e_n^3)}$$

$$e_{n+1} = \frac{\frac{1}{3}f''(\alpha)e_n^3 + O(e_n^4)}{f'(\alpha) + \frac{1}{2}f''(\alpha)e_n^2 + O(e_n^3)} = Ce_n^3 + O(e_n^4)$$

2d) 51 cubic convergence

3) Apply Newtons method with $x_0 = 1/2$ to find the root of the following functions

a) $f(x) = \sin(x)\sinh(x)$

$f'(x) = \cos(x)\sinh(x) + \sin(x)\cosh(x)$

converges to $x=0$ in 13 iterations

b) $f(x) = x - \sin(x)$

$f'(x) = 1 - \cos(x)$

Converges to $x=0$ in 12 iterations

c) $f(x) = x\cos(x)$
$f'(x) = \cos(x) - x\sin(x)$

converges to $x=0$ in 4 iterations